

Advertiser Demand Decision Lab

An independent synthetic case study in campaign-planning analytics

Brandon Crain | AI-assisted portfolio demonstration | October 2026

All advertisers and events are fictional. No Netflix, employer or client data. Not a production deployment, causal study or advertising-employment claim.

The decision

Investigate planning completion first in the default fixture. Of 300 usable current attempts, 120 complete a plan and 96 launch. Baseline counts are 210 and 150. The planning drop-off rises 30 percentage points under a shared denominator; validate the reason before choosing an intervention.

The priority is deliberately conditional. At 50% recovery assumptions, stage-specific diagnostic scores are 30 for planning and 4 for launch follow-through per 100 eligible attempts. They are not predicted launches or equal business value. Planning recovery near 6.7% equalizes those scores when assumed launch recovery is 50%.

A recommendation that can change

Fixture	Raw launch rate baseline to current	Adjusted current	Review outcome
Planning bottleneck	50% to 32%	32%	Investigate planning
Launch bottleneck	50% to 18%	18%	Investigate launch
Mix shift only	50% to 35.3%	50%	No adjusted deterioration
Incomplete telemetry	Coverage unreliable	Gate applies	Hold for measurement
No observations	Unavailable	Unavailable	Hold; no invented rate

The mix-shift fixture preserves segment outcome rates. Its raw decline is composition, not within-segment deterioration. A current-stage investigation priority is a different question from explaining the historical decline.

Try the offline demo

Open `output/dashboard.html`. Change scenario, segment and recovery assumptions; compare raw and adjusted rates, inspect exclusions, then export a cohort CSV and decision JSON. The five scenarios offer 500 precomputed views. No accounts, server or network are required.

Defensible before sophisticated

Measurement and engineering

One advertiser initial planning attempt is the analytical unit. Both cohorts have the same inclusive 14-day horizon. All funnel rates use mature, complete and valid attempts; missing telemetry is an exclusion, not a failed conversion. Baseline segment weights define the same comparison population across periods.

Python validates clocks and identities before SQLite derives one row per attempt. Exact retries collapse; event conflicts quarantine affected advertisers. Ambiguous advertiser dimensions abort the run rather than bias segment results by input order. Fractional-second timestamps are rejected instead of rounded across boundaries.

Missing segment support, small cells or excessive incomplete/invalid coverage block the recommendation. Recovery fractions and decision thresholds are illustrative policy. Wilson bounds only illustrate raw-rate binomial precision; there is no claimed causal or adjusted-rate uncertainty interval.

What improves on the prior lab

The separate entry adds an advertiser planning/launch question, temporal cohorts, a pure-mix scenario, explicit recovery sensitivity, a break-even point, coverage holds and no-data behavior. It reuses the Content Measurement Lab's strengths in grain, event-time validation, SQL lineage and interpretation. The original project is preserved.

Validation and remaining limits

The review packet includes independent mathematical and engineering reviews, deterministic rebuild checks, adversarial tests and actual-browser checks of every advertised control combination and exports. Consult docs/VALIDATION.md for the final observed counts and scope.

This is not an ad auction, demand forecast, production ML system or completed experiment. It omits budgets, supply constraints, seasonality, repeat attempts, intervention costs and revenue. Exact fixture outcomes are designed rather than fitted. A recovered completed plan does not guarantee a launch.

Before a real intervention

Reconcile instrumentation, inspect excluded advertisers, investigate abandonment reasons, estimate costs and define a common terminal outcome. Then prospectively evaluate the intervention with a primary metric and guardrails. Adding a complex model before these steps would hide uncertainty rather than resolve it.

Owner review

Explain the grain, fixed reference weights, exclusion policy, mix-shift result and score-unit limitation. Demonstrate a changed recommendation and a measurement hold. Review both exports before presenting the artifact as portfolio competence.

Rebuild: `python -B build.py`

Tests: `python -B -m unittest discover -s tests -v`

All runtime analysis uses the Python standard library. PDF generation is optional.